```
import tensorflow as tf
from tensorflow.keras import layers
import matplotlib.pyplot as plt
import numpy as np

(x_train, _), (_, _) = tf.keras.datasets.mnist.load_data()
x_train = (x_train.astype("float32") - 127.5) / 127.5
x_train = np.expand_dims(x_train, axis=-1)

BUFFER_SIZE = 60000
BATCH_SIZE = 256

dataset = tf.data.Dataset.from_tensor_slices(x_train)
dataset = dataset.shuffle(BUFFER_SIZE).batch(BATCH_SIZE)

def build_generator():
    model = tf.keras.Sequential()
    model.add(layers.Dense(7 * 7 * 256, use_bias=False, input_shape=(100,)))
    model.add(layers.BatchNormalization())
    model.add(layers.LeakyReLU())
    model.add(layers.Reshape((7, 7, 256)))
    model.add(layers.Conv2DTranspose(128, (5, 5), strides=(1, 1), padding="same", use_bias=False))
    model.add(layers.BatchNormalization())
    model.add(layers.LeakyReLU())
    model.add(layers.Conv2DTranspose(64, (5, 5), strides=(2, 2), padding="same", use_bias=False))
    model.add(layers.BatchNormalization())
    model.add(layers.LeakyReLU())
    model.add(layers.Conv2DTranspose(1, (5, 5), strides=(2, 2), padding="same", use_bias=False, activation="tanh"))
    return model

def build_discriminator():
    model = tf.keras.Sequential()
    model.add(layers.Conv2D(64, (5, 5), strides=(2, 2), padding="same", input_shape=[28, 28, 1]))
    model.add(layers.LeakyReLU())
    model.add(layers.Dropout(0.3))
    model.add(layers.Conv2D(128, (5, 5), strides=(2, 2), padding="same"))
    model.add(layers.LeakyReLU())
    model.add(layers.Dropout(0.3))
    model.add(layers.Flatten())
    model.add(layers.Dense(1, activation="sigmoid"))
    return model

generator = build_generator()
discriminator = build_discriminator()

loss_fn = tf.keras.losses.BinaryCrossentropy()

def discriminator_loss(real_output, fake_output):
    real_loss = loss_fn(tf.ones_like(real_output), real_output)
    fake_loss = loss_fn(tf.zeros_like(fake_output), fake_output)
    return real_loss + fake_loss

def generator_loss(fake_output):
    return loss_fn(tf.ones_like(fake_output), fake_output)

generator_optimizer = tf.keras.optimizers.Adam(1e-4)
discriminator_optimizer = tf.keras.optimizers.Adam(1e-4)

@tf.function
def train_step(images):
    noise = tf.random.normal([BATCH_SIZE, 100])
    with tf.GradientTape() as gen_tape, tf.GradientTape() as disc_tape:
        generated_images = generator(noise, training=True)
        real_output = discriminator(images, training=True)
        fake_output = discriminator(generated_images, training=True)
        gen_loss = generator_loss(fake_output)
        disc_loss = discriminator_loss(real_output, fake_output)
    gradients_of_generator = gen_tape.gradient(gen_loss, generator.trainable_variables)
    gradients_of_discriminator = disc_tape.gradient(disc_loss, discriminator.trainable_variables)
    generator_optimizer.apply_gradients(zip(gradients_of_generator, generator.trainable_variables))
    discriminator_optimizer.apply_gradients(zip(gradients_of_discriminator, discriminator.trainable_variables))

def generate_images(model, epoch):
    noise = tf.random.normal([16, 100])
```

```
    images = model(noise, training=False)
    fig = plt.figure(figsize=(4, 4))
    for i in range(16):
        plt.subplot(4, 4, i + 1)
        plt.imshow(images[i, :, :, 0], cmap="gray")
        plt.axis("off")
    plt.show()

EPOCHS = 10

for epoch in range(EPOCHS):
    for image_batch in dataset:
        train_step(image_batch)
    generate_images(generator, epoch + 1)
```

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  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shap
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
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